PROJECTIVE ALGEBRAIC VARIETIES

In this lecture, we will cover some basic theory of projective algebraic varieties. We will not prove all the things covered here as this and the next few lectures are to be taken more like a glossary of terms and results that we will use. The students are expected to work out the details on their own, as this will give them much needed experience with these objects.

1. Projective space and projective algebraic sets

Definition 1.1. Let k be any field. By $\mathbb{P}^n(k)$, or simply $\mathbb{P}^n$ if k is understood, we will mean the set of all $(n+1)$-tuples $(x_0, \ldots, x_n) \in \mathbb{A}^{n+1}$ such that at least one x_i is nonzero, modulo the equivalence relation $(x_0, \ldots, x_n) \sim (y_0, \ldots, y_n)$ if there exists a $\lambda \in k^*$ such that $x_i = \lambda y_i$ for all i. An equivalence class $\{(\lambda x_0, \ldots, \lambda x_n)\}$ is denoted by $[x_0, \ldots, x_n]$, these equivalence classes will be called 'points' in the projective space. $(x_0, \ldots, x_n)$ are called *homogeneous coordinates* for the corresponding point in $\mathbb{P}^n(k)$. Note that the i-th coordinate x_i is not well-defined in general, but it is a well-defined notion to say whether the i-th coordinate is 0 or not, and if $x_i \neq 0$ for some i, then the ratios x_j/x_i are well-defined.

For each fixed choice of i, we let $U_i := \{[x_0, \ldots, x_n] \in \mathbb{P}^n : x_i \neq 0\}$. Each $P \in U_i$ can be written uniquely as $P = [x_0, x_1, \ldots, x_{i-1}, 1, x_{i+1}, \ldots, x_n]$. The coordinates $(x_0, x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n)$ are called the *nonhomogeneous coordinates* for P with respect to U_i(or x_i or i). $\phi_i : (x_0, x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n) \leftrightarrow [x_0, x_1, \ldots, x_{i-1}, 1, x_{i+1}, \ldots, x_n]$ is a one-to-one correspondence between the points of $\mathbb{A}^n$ and U_i. Note that $\mathbb{P}^n$ equals the union of all U_i each of which looks like the affine n-space.

For convenience and some notational reasons, usually we will concentrate on U_n. In particular, let H_∞ denote $\mathbb{P}^n - U_n$, i.e.

$$H_\infty = \{[x_0, \ldots, x_n] : x_n = 0\}.$$

This is usually called the *hyperplane at infinity* and it may be identified with $\mathbb{P}^{n-1}$. Thus $\mathbb{P}^n = U_n \sqcup H_\infty$ is the union of an affine n-space and a set that gives all directions in the affine n-space.

Now we give the definition of a projective algebraic set. For the remainder of this section, assume that k is algebraically closed. More generally, if k is not algebraically closed, projective zero loci should be considered inside $\mathbb{P}^n(\bar{k})$, and the set of k-rational points is obtained by intersecting with $\mathbb{P}^n(k)$. We will return to this point later.

A polynomial cannot in general be evaluated at a projective point, because its value may depend on the choice of homogeneous coordinates. For a homogeneous polynomial F of degree d, however, we have

$$F(\lambda x_0, \ldots, \lambda x_n) = \lambda^d F(x_0, \ldots, x_n).$$

Consequently, although the value $F(P)$ is not intrinsically defined, the condition $F(P) = 0$ is well-defined. This motivates the use of homogeneous polynomials in projective geometry.

Definition 1.2. Let $F \in k[x_0, \ldots, x_n]$ denote a homogeneous polynomial of degree d. An ideal I of $k[x_0, \ldots, x_n]$ is called a *homogeneous ideal* if it is generated by homogeneous polynomials.

To each homogeneous ideal I, we associate a subset $V(I)$ of $\mathbb{P}^n$ by

$$V(I) = \{P \in \mathbb{P}^n : f(P) = 0 \text{ for all homogeneous } f \in I\}.$$

A *projective algebraic set* is any set of the form $V(I)$ for a homogeneous ideal I. If V is a projective algebraic set, the *homogeneous ideal of* V, denoted $I(V)$, is the ideal of $k[x_0, \ldots, x_n]$ generated by

$$\{f \in k[x_0, \ldots, x_n] : f \text{ is homogeneous and } f(P) = 0 \text{ for all } P \in V\}.$$

Note that most of the properties that arise from first principles for affine algebraic sets hold true for projective algebraic sets as well in an analogous way. The reader is encouraged to state and prove the projective analogues of all these properties. This is highly recommended. In particular, projective algebraic varieties, some of which will be the objects of our focus for this course, are defined in exactly the same way.

Definition 1.3. A nonempty projective algebraic set $V \subset \mathbb{P}^n$ is *reducible* if there exist two algebraic sets V_1 and V_2 in $\mathbb{P}^n$ and $V_i \neq V, i = 1, 2$ such that $V = V_1 \cup V_2$. Otherwise V is *irreducible*.

In particular, projective varieties are defined over algebraically closed fields in an identical way through the correspondence between irreducible sets and prime ideals.

Definition 1.4. Assume k is algebraically closed. Then a projective algebraic set V is called a projective variety if the homogeneous ideal $I(V)$ is a prime ideal in $k[x_0, \ldots, x_n]$.

For objects appearing in the projective setting, there is a possibility of considering them in the corresponding affine situation as well. (Since every homogeneous ideal is after all an ideal, for instance.) If there is any danger of confusion, we write V_a, I_a for the affine objects and V_p, I_p for the projective ones.

Definition 1.5. If V is an algebraic set in $\mathbb{P}^n$, we write

$$C(V) := \{(x_0, \ldots, x_n) \in \mathbb{A}^{n+1} : [x_0, \ldots, x_n] \in V \text{ or} (x_0, \ldots, x_n) = (0, \ldots, 0)\}.$$

This is called the *cone over* V.

If $V \neq \emptyset$, then $I_a(C(V)) = I_p(V)$; and if $I \subset k[x_0, \ldots, x_n]$ is a homogeneous ideal such that $V_p(I) \neq \emptyset$, then $C(V_p(I)) = V_a(I)$. Thus, this construction reduces many questions about $\mathbb{P}^n$ to questions about $\mathbb{A}^{n+1}$. The best example of this might be the following:

Theorem 1.6 (Projective Nullstellensatz)**.** *Assume that k is algebraically closed. Let*

$$\mathfrak{m} = (x_0, \ldots, x_n)$$

be the irrelevant ideal, and let I be a homogeneous ideal in $k[x_0, \ldots, x_n]$. Then:

(1) $V_p(I) = \emptyset$ if and only if there exists an integer $N \geq 1$ such that $\mathfrak{m}^N \subset I$.
(2) If $V_p(I) \neq \emptyset$, then $I_p(V_p(I)) = \mathrm{Rad}(I)$.

In particular, note that if I is a homogeneous ideal, then so is $\mathrm{Rad}(I)$. (This needs proof before going to the Nullstellensatz.) You should be able to prove the above theorem from the affine Nullstellensatz. In particular, there is an inclusion-reversing one-to-one correspondence between nonempty projective algebraic sets and homogeneous radical ideals I satisfying $I \subsetneq \mathfrak{m}$. Under this correspondence, projective varieties correspond to homogeneous prime ideals $\mathfrak{p}$ satisfying $\mathfrak{p} \subsetneq \mathfrak{m}$. The exceptional homogeneous prime ideal $\mathfrak{m}$ corresponds to the empty projective algebraic set. You are encouraged to write all of these statements down precisely. In particular, reduced projective hypersurfaces $V = V(F)$ correspond to nonconstant squarefree forms F, up to multiplication by a nonzero scalar. Irreducible projective hypersurfaces correspond to irreducible forms, again up to multiplication by a nonzero scalar. (Note that a form is the same as a homogeneous polynomial.) A *hyperplane* is a hypersurface defined by a form of degree one.

The correspondence between projective and affine situations is also understood through the process of *homogenization* and *dehomogenization*. Let V be a projective algebraic set with homogeneous ideal $I(V)$. Then for any fixed index i between 0 and n, we can consider $\phi_i^{-1}(V \cap U_i)$. Introduce affine coordinates $y_j = x_j/x_i$ for $j \neq i$. For $F \in k[x_0, \ldots, x_n]$, define its dehomogenization with respect to x_i by

$$F^{(i)}(y_0, \ldots, \widehat{y_i}, \ldots, y_n) = F(y_0, \ldots, y_{i-1}, 1, y_{i+1}, \ldots, y_n).$$

Then $\phi_i^{-1}(V \cap U_i)$ is an affine algebraic set with ideal

$$I\big(\phi_i^{-1}(V \cap U_i)\big) = \{F^{(i)} : F \in I(V)\} \subset k[y_0, \ldots, \widehat{y_i}, \ldots, y_n].$$

Note that

$$V = (V \cap U_0) \bigcup \cdots \bigcup (V \cap U_n)$$

shows that V is covered by these affine algebraic subsets via the appropriate maps ϕ_i^{-1}.

The above process can be reversed. For any

$$f \in k[y_0, \ldots, \widehat{y_i}, \ldots, y_n],$$

define

$$f^*(x_0, \ldots, x_n) := x_i^d f\left(\frac{x_0}{x_i}, \ldots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_n}{x_i}\right),$$

where $d = \deg(f)$ is the smallest integer for which f^* is a polynomial. We say that f^* is the *homogenization of f with respect to x_i*.

Definition 1.7. Let $V \subset \mathbb{A}^n$ be an affine algebraic set with ideal $I(V)$ and consider V as a subset of $\mathbb{P}^n$ via ϕ_i for some fixed i. The *projective closure* of V, denoted by $\bar{V}$, is the projective algebraic set whose homogeneous ideal $I(\bar{V})$ is generated by $\{f^* : f \in I(V)\}$. Equivalently, $\bar{V}$ is the smallest projective algebraic set containing V.

Remark 1.8. In defining the projective closure, it is important that we homogenize all the elements of $I(V)$. Homogenizing only an arbitrary generating set for $I(V)$ can introduce extra components at infinity. Equivalently, one may homogenize a generating set and then saturate the resulting homogeneous ideal with respect to x_i. For a hypersurface, homogenizing its single defining equation is sufficient.

Proposition 1.9. *We give the correspondence between an affine variety and its projective closure.*

(1) Let V be an affine variety, considered as a subset of $U_i \simeq \mathbb{A}^n$. Then $\bar{V}$ is a projective variety and $V = \bar{V} \cap U_i$ under this identification.

(2) Let V be a projective variety. Then $V \cap U_i$ is either empty or, under the identification $U_i \simeq \mathbb{A}^n$, it is an affine variety. In the nonempty case,

$$V = \overline{V \cap U_i}.$$

The proposition above is the reason many times we will say "let V be a projective variety" and write down some inhomogeneous equation with the understanding that we are talking about the projective closure of the affine variety defined by that equation. The points in $\bar{V} \setminus V$ are called the *points at infinity* on V.

2. Theory of projective algebraic sets over non-algebraically closed fields

We will be brief in this section, since most notions are straightforward analogues of the same notions in the affine case. Let K be a perfect field and $\bar{K}$ be a fixed algebraic closure of K with $G_K = \mathrm{Gal}(\bar{K}/K)$ as before. If $I \subset K[x_0, \ldots, x_n]$ is a homogeneous ideal, its geometric projective zero locus is considered inside $\mathbb{P}^n(\bar{K})$, and its set of K-rational points is

$$V(K) = V(\bar{K}) \cap \mathbb{P}^n(K).$$

The action of the Galois group on $\mathbb{P}^n(\bar{K})$ is well-defined coordinatewise. It follows that $\mathbb{P}^n(K)$ is again the set of fixed points of $\mathbb{P}^n(\bar{K})$ under this Galois action. Similarly, the Galois action restricts to $V(\bar{K})$ for any projective algebraic set V defined over K and we can retrieve the set $V(K)$ as the set of fixed points under the Galois action.

If $P = [x_0, \ldots, x_n] \in \mathbb{P}^n(K)$ and the displayed coordinates are chosen in $\bar{K}$, the individual x_i need not lie in K. Nevertheless, there exists $\lambda \in \bar{K}^*$ such that $\lambda x_0, \ldots, \lambda x_n \in K$. Equivalently, choosing some i with $x_i \neq 0$, we have $x_j/x_i \in K$ for every j. In general, we have the following definition.

Definition 2.1. Let $P = [x_0, \ldots, x_n] \in \mathbb{P}^n(\bar{K})$. The minimal field of definition for P over K is the field

$$K(P) = K(x_0/x_i, \ldots, x_n/x_i)$$

for any i with $x_i \neq 0$.

It follows from above that $K(P)$ equals the fixed field of $\{\sigma \in G_K : P^\sigma = P\}$.

As before, we want to understand functions on a projective variety V. The definitions are straightforward analogues as before.

Definition 2.2. Let V be a nonempty projective variety defined over K, i.e. V is defined over K and V is an irreducible projective algebraic set over $\bar{K}$. Then the homogeneous coordinate ring of V over K is defined to be

$$K_h[V] = \frac{K[x_0, \ldots, x_n]}{I(V/K)},$$

where $I(V/K)$ is defined analogously as before. This is an integral domain, and it inherits a grading from $K[x_0, \ldots, x_n]$. We denote its fraction field by

$$K_h(V) = \mathrm{Frac}(K_h[V]).$$

We can similarly define $\bar{K}_h[V]$ and $\bar{K}_h(V)$ by replacing K by $\bar{K}$ and $I(V/K)$ by $I(V)$.

Unlike the case of affine varieties, no homogeneous elements of positive degree in $K_h[V]$ or $\bar{K}_h[V]$ determine functions on V. Similarly, most elements of $K_h(V)$ or $\bar{K}_h(V)$ cannot be regarded as functions on V. However, if f and g are homogeneous elements of $K_h[V]$ of the same degree and $g \neq 0$, then f/g defines a rational function on V. It is regular on the open subset

$$D_+(g) = \{P \in V : g(P) \neq 0\}.$$

(In fact, degrees of numerator and denominator being different is the reason why most other fractions cannot be well-defined functions on V.) This is how we define the function field of V.

Definition 2.3. The function field of V over K, denoted $K(V)$, is the degree-zero subfield of $K_h(V)$:

$$K(V) = K_h(V)_0 = \left\{ \frac{f}{g} : f, g \in K_h[V] \text{ are homogeneous of the same degree and } g \neq 0 \right\}.$$

We similarly define $\bar{K}(V)$ by replacing K by $\bar{K}$. Elements of the function field are called *rational functions* on V.

Note that $K \subset K(V) \subset K_h(V)$ but $K_h[V] \not\subset K(V)$. Now we define the local ring at a point.

Definition 2.4. Let V be a projective algebraic variety, let $P \in V(\bar{K})$, and let $z \in K(V)$. We say that z is defined at P if z can be written as f/g with f, g homogeneous elements of the same degree and $g(P) \neq 0$. We define

$$\mathcal{O}_{V,P} := \{z \in K(V) : z \text{ is defined at } P\}.$$

This is a subring of $K(V)$. It is a local ring with maximal ideal

$$\mathfrak{m}_{V,P} := \{z \in \mathcal{O}_{V,P} : z(P) = 0\}.$$

This is called the *local ring of V at P*. Its elements are also called *regular functions at P*. The value $z(P)$ of a function $z \in \mathcal{O}_{V,P}$ is well-defined. Moreover, the residue field

$$\mathcal{O}_{V,P}/\mathfrak{m}_{V,P}$$

is naturally isomorphic to $K(P)$.

We can define corresponding objects over $\bar{K}$ in the same manner. Does the Galois action work in an analogous way as well? Keep in mind that most of the times we would be working with the $\bar{K}$ versions of these objects.

3. Dimension and smoothness

In this section, we describe further properties of projective varieties such as dimension and smoothness. We can make use of the affine theory to make these definitions.

Definition 3.1. Let V be a projective variety over K and choose an affine chart $U_i \simeq \mathbb{A}^n$ such that $V \cap U_i \neq \emptyset$. The dimension of V is defined by

$$\dim V = \dim(V \cap U_i).$$

This is independent of the choice of i, since every nonempty open subset of an irreducible variety has the same dimension. Equivalently,

$$\dim V = \operatorname{trdeg}_K K(V).$$

In particular, if $K_h[V]$ is the homogeneous coordinate ring of V, then

$$\dim K_h[V] = \dim V + 1.$$

Definition 3.2. Let V be a projective variety over K, let $P \in V$, and choose an affine chart $U_i \simeq \mathbb{A}^n$ such that $P \in V \cap U_i$. Then V is nonsingular (or smooth) at P if $V \cap U_i$ is nonsingular (or smooth) at P.

This definition is independent of the choice of affine chart, since

$$\mathcal{O}_{V,P} \simeq \mathcal{O}_{V \cap U_i, P}$$

whenever $P \in U_i$, and smoothness is a local property.

Proposition 3.3 (Projective Jacobian criterion)**.** *Let*

$$V = V(F_1, \ldots, F_m) \subset \mathbb{P}^n$$

be a projective variety of dimension r, where the F_i are homogeneous generators of $I(V)$. A point $P \in V$ is smooth if and only if

$$\operatorname{rank} \left(\frac{\partial F_i}{\partial x_j}(P) \right)_{\substack{1 \le i \le m \\ 0 \le j \le n}} = n - r.$$

In particular, if $V = V(F)$ is a projective hypersurface, then $P \in V$ is singular if and only if

$$\frac{\partial F}{\partial x_0}(P) = \cdots = \frac{\partial F}{\partial x_n}(P) = 0.$$

Remark 3.4. In fact, it turns out that we can define the function field and rings of functions at points also in terms of these affine covers. We leave it for the students to check that these definitions are equivalent.

Proposition 3.5. *Let V be a projective variety over K and choose an affine chart $U_i \simeq \mathbb{A}^n$ such that $V \cap U_i \neq \emptyset$. Then restriction induces a canonical isomorphism*

$$K(V) \simeq K(V \cap U_i),$$

and similarly over $\bar{K}$. If $P \in V \cap U_i$, then there is a canonical isomorphism

$$\mathcal{O}_{V,P} \simeq \mathcal{O}_{V \cap U_i, P},$$

and similarly over $\bar{K}$.